Evaluating and Improving the Reliability of a FinRobot Equity Research Agent

GR5398 — Assignment 1 | Fall 2026
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Study universe: NVIDIA (NVDA), Advanced Micro Devices (AMD), Intel (INTC), Broadcom (AVGO), Qualcomm (QCOM)
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Abstract

This project implements a multi-company equity research workflow with FinRobot and examines whether financial narratives generated by a large language model are numerically grounded, reproducible, and stable. Five semiconductor companies are analyzed using financial tables, forecasts, peer comparisons, machine-generated investment narratives, and HTML research reports. For reliability assessment, 22 financial records for AVGO and QCOM are reconciled to locally saved SEC Company Facts; two illustrative five-year discounted cash-flow (DCF) scenarios are constructed; 206 candidate statements across three companies are preliminarily audited for evidentiary grounding; three independent NVDA narrative runs are compared; and a financial grounding verifier is revised and tested against a baseline. On a curated 14-passage benchmark, passage-level verifier F1 rises from 0.667 to 1.000. The result demonstrates improvement on the evaluated cases, not production-level generalization. The full candidate-sentence grounding audit found 113/206 Supported statements (54.85%), with particularly low grounding in risk narratives. The main takeaway is that deterministic reconciliation and explicit provenance are important complements to LLM-generated financial prose.

1. Research Objectives

Financial equity research combines structured accounting data with qualitative interpretation. An LLM can produce readable analysis quickly, but errors may arise when it mixes fiscal periods, conflates actuals with forecasts, references unsupported valuation figures, or compares incompatible metrics. This assignment addresses three questions:

1. Can FinRobot generate research deliverables for five related semiconductor companies using heterogeneous financial-data sources?
2. How reliable and stable are its financial claims when compared with recorded evidence and across repeated generation runs?
3. Does a targeted, rule-based financial grounding verifier reduce known numerical errors without introducing excessive false positives?

The study evaluates financial-text quality, not investment performance. Nothing herein should be interpreted as a live trading recommendation.

2. Implementation and Data Provenance

The five-company universe consists of NVDA, AMD, INTC, AVGO, and QCOM. Historical records and illustrative forecasts were processed into normalized financial tables and peer comparisons, and seven narrative sections were generated for each company. Five professional HTML equity reports were produced and passed 25 internal technical checks (five checks per company) for the expected sections and data consistency after a peer-comparison correction.

The original workflow uses Financial Modeling Prep (FMP) where accessible. Because the available FMP Free endpoints did not provide the needed AVGO and QCOM statements, these two companies use locally saved SEC EDGAR Company Facts and XBRL concepts. This distinction is retained in the research provenance. SEC Company Facts is a primary regulatory data source, but selecting the appropriate taxonomy concept, fiscal period, and financial definition remains an analyst responsibility. Financial projections are scenario outputs, not company guidance.

A material problem identified during the workflow was an inconsistent NVDA peer comparison: one earlier 2027E EBITDA figure was approximately $382 billion even though the main NVDA projection was approximately $159.85 billion. The peer table was rebuilt from the same per-company analysis files used elsewhere in the report. The original failure became a useful positive test case for the verifier improvement.

3. Part 1: Multi-Company Equity Research

Five HTML research reports and corresponding financial/forecast tables were generated. This demonstrates end-to-end execution across a coherent peer universe. However, a completed HTML artifact is not equivalent to a verified investment recommendation. The reports retain methodological limitations, particularly where source definitions differ or prospective margins imply aggressive assumptions.

For context, the model inputs included NVDA FY2026 revenue of $215.938 billion and an EBITDA figure of $144.552 billion, with projected 2027E EBITDA around $159.848 billion. AMD's model indicated a substantial jump in projected EBITDA margin from the FY2025 level into FY2026; that result is a scenario assumption requiring independent economic justification rather than an established fact. Comparison across semiconductor companies is affected by different fiscal year-ends, business models, and accounting presentation.

4. Part 2A: Financial Data Reconciliation

The audit evaluated 11 FY2025 records each for AVGO and QCOM. Nine direct records per company were linked to an identified SEC XBRL concept; two derived records per company were independently recomputed from selected component values. All 22 tested records matched the locally saved SEC Company Facts at the script's USD tolerance.

Company	Direct	Derived	Matched	Mismatch	Missing
AVGO	9	2	11	0	0

Company	Direct	Derived	Matched	Mismatch	Missing
QCOM	9	2	11	0	0
Total	18	4	22	0	0

The derived measures are labeled **FCF proxy = operating cash flow minus capital expenditure** and **EBITDA proxy = operating income plus selected depreciation/amortization components**. AVGO FY2025 FCF proxy is $26.914 billion and EBITDA proxy is $34.120 billion. QCOM FY2025 FCF proxy is $12.820 billion and EBITDA proxy is $13.957 billion. The AVGO SEC filing accession was `0001730168-25-000121`; the QCOM accession was `0000804328-25-000085`.

Interpretation constraint: The 22/22 result establishes *internal consistency with the selected SEC XBRL facts*, not independent agreement with every line of the original 10-K narrative. The normalized CSV and comparison values share an underlying SEC source. In particular, QCOM capital expenditure classification (`PaymentsToAcquireProductiveAssets`) and AVGO depreciation/amortization composition warrant direct filing review before being presented as definitive reported metrics. The count and scope of the audit should not be conflated with independent financial-statement assurance.

5. Part 2B: Illustrative Discounted Cash-Flow Analysis

Using the audited FY2025 FCF proxies as starting values, a five-year cash-flow model projected 2026–2030 growth at 5%, 6%, 4%, 4%, and 4%, respectively. A base discount rate of 11% and terminal growth of 3% produced the following discounted cash-flow proxies:

Company	FY2025 FCF proxy	Base discounted cash-flow proxy	Sensitivity range
AVGO	$26.914B	$371.348B	$277.80B–$575.56B
QCOM	$12.820B	$176.885B	$132.33B–$274.16B

The sensitivity grid varies the discount rate from 9% to 13% and terminal growth from 2% to 4%. Values fall as the discount rate increases and rise as terminal growth increases, confirming the expected direction of model sensitivity. Both outputs change substantially over the chosen grid, showing why reporting a single point estimate without scenarios can be misleading.

Financial interpretation constraint: Because operating cash flow minus capex is an FCF proxy, the discount rate is an illustrative rather than calibrated WACC, and net debt/cash/share-count adjustments were not completed, these values are *cash-flow present-value scenarios*. They are neither verified enterprise values nor equity values and **must not** be called per-share targets. The assumptions are pedagogical and not claimed to be externally estimated capital-market parameters.

6. Part 2C: Evidence Grounding Audit

6.1 Scope and methodology

The expanded audit reviewed **206 candidate narrative statements** from NVIDIA (NVDA), AMD, and Broadcom (AVGO), spanning seven substantive FinRobot-generated sections per company.

Each candidate statement received a preliminary whole-sentence label:

- **Supported:** its material factual and numerical content agrees with the evidence available to the generating agent.
- **Unsupported:** relevant evidence is absent or insufficient, including unevidenced causal or market interpretations.
- **Contradicted:** a material claim conflicts with the recorded evidence, such as a wrong peer-company amount or an incorrect Actual/Estimate period designation.

Grounding Rate is defined as the number of Supported candidate statements divided by the total number of reviewed candidates. The audit combines automated evidence assistance and manual-style review. Labels represent a preliminary sentence-level adjudication, not an independently validated atomic-claim gold standard.

6.2 Overall and company-level results

Of the 206 candidate statements, **113 were Supported, 66 Unsupported, and 27 Contradicted**, yielding an overall candidate-level Grounding Rate of **54.85%**.

Company	Total	Supported	Unsupported	Contradicted	Grounding Rate
AMD	72	33	30	9	45.83%
AVGO	67	49	11	7	73.13%
NVDA	67	31	25	11	46.27%

The difference between company rates is descriptive rather than causal; company-specific claim composition and evidence coverage may influence the results.

6.3 Agent-section results

Agent Section	Total	Supported	Unsupported	Contradicted	Grounding Rate
company_overview	48	23	20	5	47.92%
competitor_analysis	35	13	11	11	37.14%
investment_overview	40	27	11	2	67.50%
major_takeaways	22	21	1	0	95.45%
risks	17	3	11	3	17.65%
tagline	9	7	1	1	77.78%
valuation_overview	35	19	11	5	54.29%

The strongest section was **major_takeaways**, where the agent primarily restated recorded growth and margin figures. The weakest was **risks**, which often extrapolated historical accounting trends into speculative business or market consequences without corresponding evidence.

6.4 Full company-by-section breakdown

Company	Agent Section	Total	Supported	Unsupported	Contradicted	Grounding Rate
AMD	company_overview	16	7	7	2	43.75%
AMD	competitor_analysis	12	3	5	4	25.00%
AMD	investment_overview	16	8	6	2	50.00%
AMD	major_takeaways	8	7	1	0	87.50%
AMD	risks	5	0	5	0	0.00%
AMD	tagline	3	3	0	0	100.00%
AMD	valuation_overview	12	5	6	1	41.67%
AVGO	company_overview	17	11	5	1	64.71%
AVGO	competitor_analysis	12	6	2	4	50.00%
AVGO	investment_overview	12	11	1	0	91.67%
AVGO	major_takeaways	6	6	0	0	100.00%
AVGO	risks	5	2	2	1	40.00%
AVGO	tagline	3	3	0	0	100.00%
AVGO	valuation_overview	12	10	1	1	83.33%
NVDA	company_overview	15	5	8	2	33.33%
NVDA	competitor_analysis	11	4	4	3	36.36%
NVDA	investment_overview	12	8	4	0	66.67%
NVDA	major_takeaways	8	8	0	0	100.00%
NVDA	risks	7	1	4	2	14.29%
NVDA	tagline	3	1	1	1	33.33%
NVDA	valuation_overview	11	4	4	3	36.36%

6.5 Observed grounding failures

Three recurring failure mechanisms were identified:

1. **Actual/Estimate confusion.** Historical values marked `2025A` or `2026A` were sometimes described as forecasted or estimated.
2. **Peer-figure attribution errors.** A generated NVDA valuation statement gave 2027E peer EBITDA estimates of $8.6B for AMD and $47.4B for INTC, whereas the saved comparison table records approximately $15.614B and $17.118B, respectively.
3. **Unsupported causal interpretation.** Correct financial figures were used to assert unverified claims about pricing power, management efficiency, investor sentiment, future demand, or business risk.

These patterns illustrate why numerical consistency alone is not equivalent to fully grounded financial analysis.

6.6 Limitations and reproducibility

The 206 observations are extracted **candidate sentences**, not necessarily all atomic propositions in the complete reports. Compound statements can mix supported figures with unsupported interpretation. Several qualitative annotations warrant further independent review. Grounding was checked relative to the saved FinRobot input evidence, not independently against all public company filings.

Seven substantive agent sections per company were evaluated. News-summary placeholders without substantive evidence-bearing statements are recorded as **N/A**, not as 100% grounded.

The original 24-claim exploratory sample remains available as a separate historical experiment but is superseded by the 206-candidate audit for section-level reporting.

Reproduction files: `experiments/part2_grounding_audit/grounding_review_worklist.csv` and `experiments/part2_grounding_audit/final_section_grounding_rates.csv`.

7. Part 2D: Three-Run NVDA Stability Test

The same frozen NVDA financial tables, model configuration, and prompt were reused across three independent runs to generate investment-overview and risk sections. The purpose was to examine narrative variation while holding recorded financial inputs fixed.

Three prominent risk themes recurred in all runs: revenue-growth volatility, elevated valuation multiples, and uncertainty around SG&A assumptions. Other risk emphasis varied (e.g., competitive pressure, margin swings, costs). Crucially, historical FY2026 actual values were described with estimated/projected language in all three runs. This supports a recurring *error pattern within this three-run experiment*, not a universal claim about the model.

The model did not produce three comparable, independently computed Buy/Hold/Sell recommendations or per-share price targets. Only Run 3 mentioned the external analyst consensus target of $339.17 with a Buy rating, information explicitly supplied in the experiment input. Thus a quantitative price-target standard deviation or independent recommendation agreement rate is **not available** and is not reported as zero. This limitation is itself a stability finding: valuation information was inconsistently surfaced and did not constitute an independent valuation.

8. Part 3: Financial Grounding Verifier Improvement

A focused rule-based verifier checks financial amounts, units, periods, and relationships against known company metrics. Version 0.1 exhibited false alarms on valid dollar figures and missed a catastrophic 2027E EBITDA mismatch. Version 0.2 improved monetary-unit normalization, detection of certain large financial mismatches, and handling of duplicated actual-versus-estimate alerts.

The same curated set of 14 passages was evaluated with both versions: 12 manually constructed examples targeting known error modes and two previously generated agent passages. The binary evaluation unit was a whole passage: a finding of one or more issues counted as an error detection. Results were:

Dataset	Version	TP	FP	FN	TN	Precision	Recall	F1
Constructed (n=12)	v0.1	4	5	1	2	0.444	0.800	0.571
Constructed (n=12)	v0.2	5	0	0	7	1.000	1.000	1.000
All available (n=14)	v0.1	6	5	1	2	0.545	0.857	0.667
All available (n=14)	v0.2	7	0	0	7	1.000	1.000	1.000

On the full benchmark, F1 improved by **33.3 percentage points**, precision by **45.5 points**, and recall by **14.3 points**. Five false positives and one false negative were eliminated on these cases. A correct $144.6B 2026A EBITDA passage illustrates an earlier false alarm; a false $382B 2027E EBITDA passage illustrates a previously missed error.

Evaluation limitation: The test set is small, manually curated, and closely aligned with error patterns used to design the improvement. The two actual Agent passages were detected by both versions. Thus the perfect v0.2 score is a *targeted regression-test result*, not evidence of perfect performance on unseen LLM reports. Future work requires a larger independently labeled sample with more companies, financial concepts, and adversarial claim styles.

Updated Part 2B — Native Valuation Engine Reconciliation

In addition to the illustrative AVGO/QCOM DCF scenarios described above, I independently reproduced NVIDIA's valuation using the original `ValuationEngine.calculate_ev_ebitda_valuation()` implementation.

The engine selected forecast-year 2029E EBITDA of $183.7151 billion. Using an explicitly specified 12.0x EV/EBITDA multiple, illustrative shares outstanding of 24.5 billion, and the engine's assumption that net debt equals 10% of enterprise value, the native engine returned an implied value of **$80.9846 per share**.

The independent arithmetic was:

- Enterprise value = $183.7151B x 12 = approximately $2,204.5815B.
- Imputed net debt = 10% of enterprise value.
- Equity value = 90% of enterprise value.
- Implied price = equity value / 24.5B shares = $80.9846.

The reconciliation difference was **$0.00**.

I also ran the original `SensitivityAnalyzer` using revenue perturbations of +/-5% and EBITDA margin changes of +/-2 percentage points. Its operating scenarios were translated into per-share values with the same EV/EBITDA pricing rule. Across 25 joint scenarios, the implied prices ranged from **$74.8419 to $87.3477**, or approximately -7.59% to +7.86% relative to the engine baseline.

These values are illustrative model outputs, **not verified price targets**. Share count and net debt are modeling assumptions. The engine's constant valuation multiple, simplified balance-sheet adjustment, and default forecast-year selection remain important limitations. The native sensitivity module generates

operating scenarios, not share-price estimates by itself; price translation was an additional deterministic calculation.

Reproduction materials: `experiments/part2_valuation/native_engine_reconciliation/` and `scripts/reconcile_native_valuation.py`.

Updated Part 3 — Verifier v0.3 and Real-Report Evaluation

The final improvement is an opt-in, post-generation, rule-based Financial Grounding Verifier integrated into the report generation pipeline. It generates JSON findings while retaining the original LLM output. It does not automatically rewrite the report text.

Version 0.3 adds local financial-metric attribution to avoid comparing explicitly labeled Revenue or Cost of Operations amounts against EBITDA. It also introduces a targeted historical-versus- forecast wording check.

On the same curated 14-passage benchmark:

Version	TP	FP	FN	TN	Precision	Recall	F1
v0.1	6	5	1	2	0.5455	0.8571	0.6667
v0.2	7	0	0	7	1.0000	1.0000	1.0000
v0.3	7	0	0	7	1.0000	1.0000	1.0000

On seven original NVDA report sections, v0.2 raised three EBITDA-mismatch alerts that manual review identified as metric-attribution false positives: one Revenue amount and two Cost of Operations amounts. Version 0.3 eliminated all three without losing detection of the constructed $382B 2027E EBITDA error.

A separate offline run completed successfully across **21 original sections** from NVDA, AMD, and AVGO. Version 0.3 returned zero findings, which means **no errors were flagged by the current rules**, not that every statement was supported or correct.

This is a measured improvement in screening specificity, not a demonstrated improvement in whole-report Grounding Rate. Automated flagging does not change original text, and a complete before/after claim-grounding comparison remains unavailable.

Reproduction materials: `experiments/part3_verifier/`, `finrobot_equity/core/src/financial_grounding_verifier.v03.py`, and `experiments/part3_verifier/finrobot_improvement_final.patch`.

Setup, Runtime and Cost Disclosure

The workflow used OpenAI-backed text generation and a combination of FMP and cached SEC data. The exact model identifier, token usage, per-company elapsed runtime, API expenditure, and total spend have not been independently reconstructed from complete execution logs. They are therefore **not**

numerically reported rather than being assigned invented values. Reproduction requires recording these quantities during future runs.

9. Synthesis and Limitations

This project demonstrates that output generation, numerical consistency, evidentiary grounding, and repeatability are distinct dimensions of an equity research agent. Generating five readable reports validates workflow execution. Matching 22 SEC-derived values validates selected extraction pathways but does not independently validate accounting semantics. DCF scenarios illuminate valuation uncertainty but should not be mistaken for stock-price recommendations. Grounding checks and stability runs reveal how apparently coherent narratives may still confuse fiscal-year labels or selectively mention external valuation figures. The verifier experiment shows concrete benefits from deterministic unit and relationship checks but is too small to establish broad accuracy.

Data access constraints also matter: FMP Free coverage varied by ticker, SEC XBRL concepts required mapping, fiscal years differed across issuers, and the report did not include a rigorous stock-price valuation based on independently estimated WACC, net debt, diluted shares, or a broad out-of-sample verifier benchmark.

10. Conclusion

FinRobot can support a multi-company research workflow efficiently, but its natural-language financial analysis should be treated as a draft requiring evidence verification. The strongest contribution is the integration of source-aware auditing, explicit scenario assumptions, repeated-generation diagnostics, and a measured verifier improvement. A future production system should verify atomic claims against primary filings, retain traceable provenance, enforce Actual/Estimate period semantics, separate analyst consensus from internal valuations, and benchmark verification against independently reviewed examples.

Reproduction and Source Notes

The submission includes five HTML reports, financial tables, SEC financial reconciliation records, DCF forecasts and sensitivity matrices, a preliminary 24-claim audit, NVDA stability outputs, verifier versions, and per-case precision/recall/F1 results. Some scripts require the larger original FinRobot source tree and locally saved SEC XBRL inputs; the lightweight submission package preserves outputs but is **not claimed to be self-contained for every experiment**.

Primary evidence sources: SEC EDGAR Company Facts for AVGO and QCOM (filing accessions as above), FMP data where available for other companies, and the locally saved experiment artifacts. All figures refer to the stated project snapshot and model assumptions.